

Tetris

1. Hardware Organization

The game will be implemented as a modular, synchronous VHDL design, organized into separate components that interact through clearly defined signals and a top-level controller.

The Top-Level Structure will consist of:

- Input Subsystem for the NES controller interface
- Game logic subsystem including the FSM, game board, and piece logic
- VGA video output
- Timing with clock dividers to establish the game speed

2. Major Hardware Modules

1. NES Controller Interface

Reads and decodes player input from NES controller.

Interface will need to:

- Generate latch/clock signals
- Decode button states
- Take serial button data
- Output debounced button signal

2. Game state controller

Control flow of the game using FSM with the following states:

- Init
- Idle
- Spawn
- Fall
- Move
- Lock
- Clear
- Game_over

State will determine when pieces move, lock in place, or disappear when a row is cleared as well as progression for score and levelling.

3. Grid Storage

Store current state of the Tetris grid handled by a 2D array/grid stored in BRAM or registers where each cell will be marked as occupied/unoccupied.

Grid storage will be able to read current board state for video output, update the board when pieces lock at the bottom, and shift blocks down when rows are cleared.

4. Piece Logic

Manages actively falling pieces. Piece logic will have to:

- Generate new pieces
- Track piece position
- Track rotation
- Output coordinates for collision logic and video rendering

5. Collision Detection

Ensure piece moves are valid by checking wall boundaries, floor boundaries, and blocking overlap with preexisting blocks.

This module will have to:

- Block illegal moves
- Output when a piece is ready to lock

6. Line Clear Logic

Detect and delete full rows. Clear logic will have to:

- Scan board for full rows
- Clear rows and shift higher rows down
- Update game score

7. VGA Output

Display game visuals, use Lab 1 code to make:

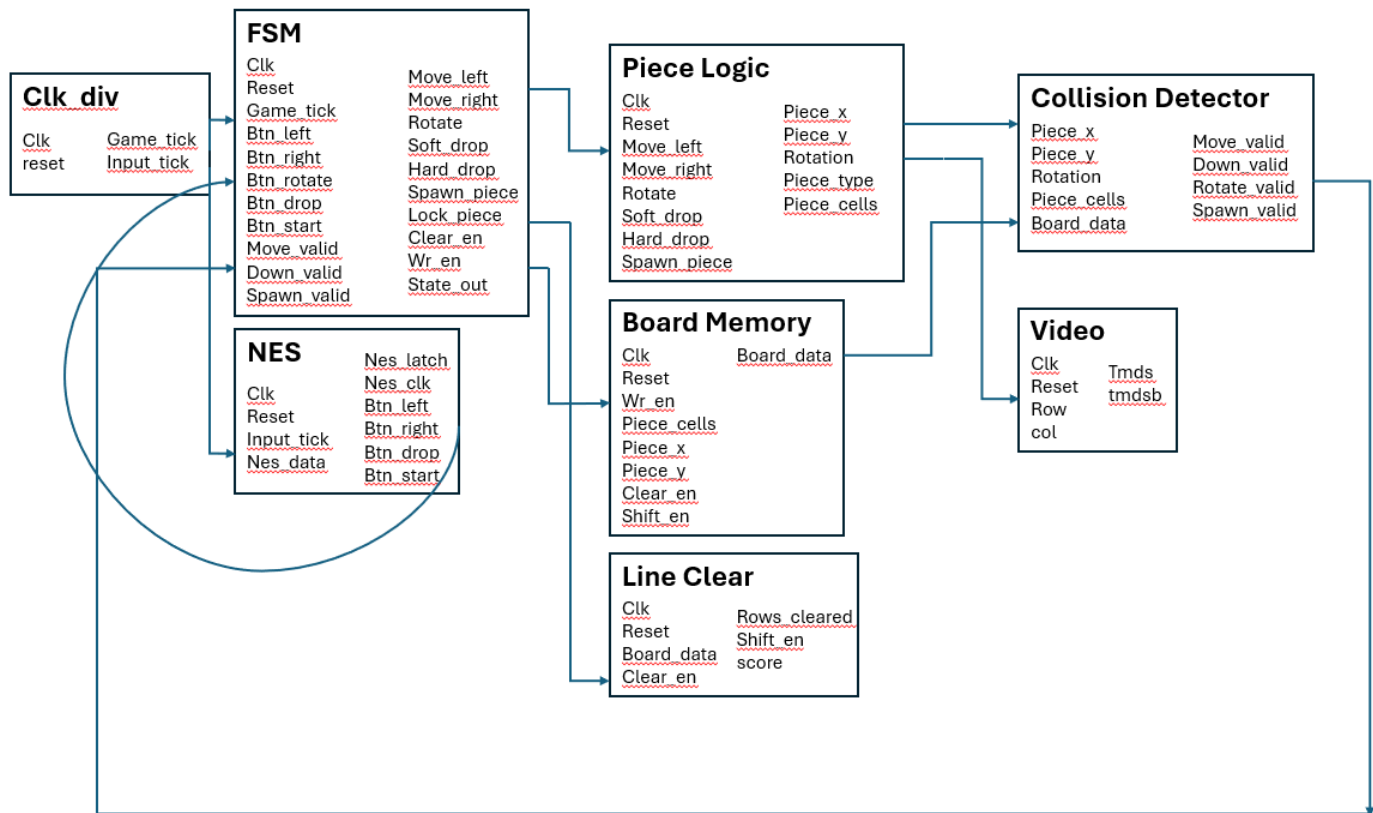
- Grid
- Active piece
- Locked pieces
- Game UI

8. Timing

Controls game speed using clock dividers to:

- Set piece falling rate
- Generate input sampling for controller

Block Design:



FSM:

